Basic Concept of Software Design

Designer's goal is to produce a model or representation of an entity that will later be built. [T]here are two major phases to any design process: diversification and convergence. Diversification is the *acquisition* of a repertoire of alternatives, the raw material of design: components, component solutions, and knowledge, all contained in catalogs, textbooks, and the mind. During convergence, the designer chooses and combines appropriate elements from this repertoire to meet the design objectives, as stated in the requirements document and as agreed to by the customer. The second phase is the gradual *elimination* of all but one particular configuration of components, and thus the creation of the final product.

Diversification and convergence combine intuition and judgment based on experience in building similar entities, a set of principles and/or heuristics that guide the way in which the model evolves, a set of criteria that enables quality to be judged, and a process of iteration that ultimately leads to a final design representation.

Software design sits at the technical kernel of software engineering and is applied regardless of the software process model that is used.

Beginning once software requirements have been analyzed and specified, ***software design is the first of three technical activities***

- ✓ design,
- ✓ code generation, and
- ✓ Test which is required to build and verify the software.

Each activity transforms information in a manner that ultimately results in validated computer software

Software Design

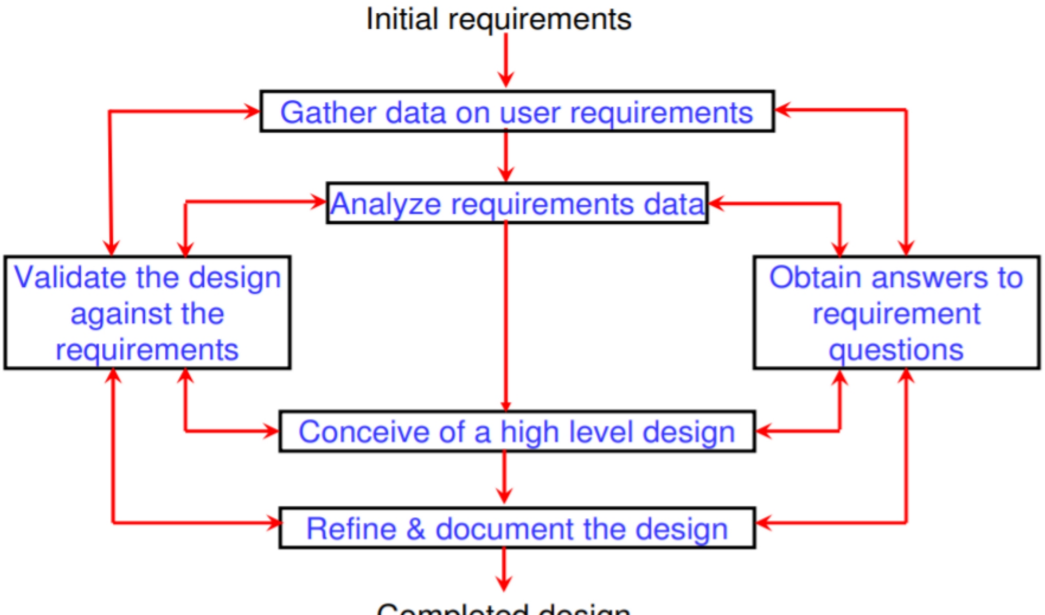

Fig. 1 : Design framework

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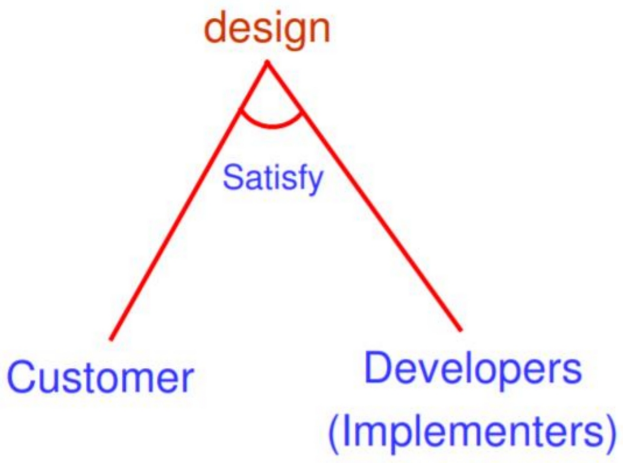
Software Design



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Conceptual Design and Technical Design

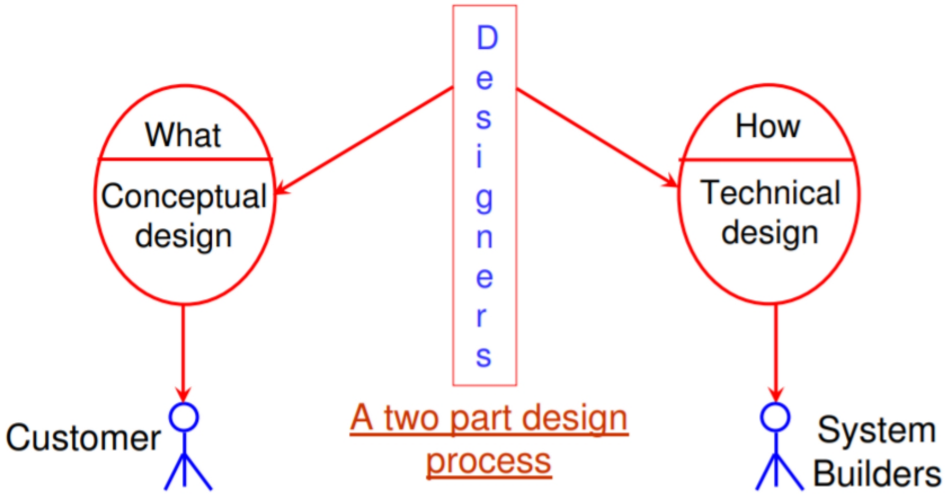

Fig. 2 : A two part design process

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Software Design

Conceptual design answers :

- ✓ Where will the data come from ?
- ✓ What will happen to data in the system?
- ✓ How will the system look to users?
- ✓ What choices will be offered to users?
- ✓ What is the timings of events?
- ✓ How will the reports & screens look like?

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Software Design

Technical design describes :

- Hardware configuration
- Software needs
- Communication interfaces
- I/O of the system
- Software architecture
- Network architecture
- Any other thing that translates the requirements in to a solution to the customer's problem.

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Software Design

The design needs to be

- Correct & complete
- Understandable
- At the right level
- Maintainable

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Software Design

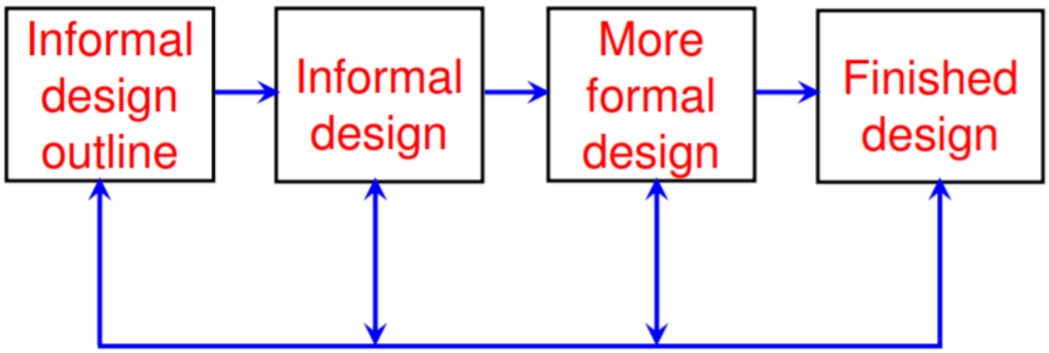


Fig. 3 : The transformation of an informal design to a detailed design.

The flow of information during software design is illustrated in Figure 13.1. Software requirements, manifested by the data, functional, and behavioral models, feed the design task. Using one of a number of design methods ,***the design task produces a data design, an architectural design, an interface design, and a component design***.

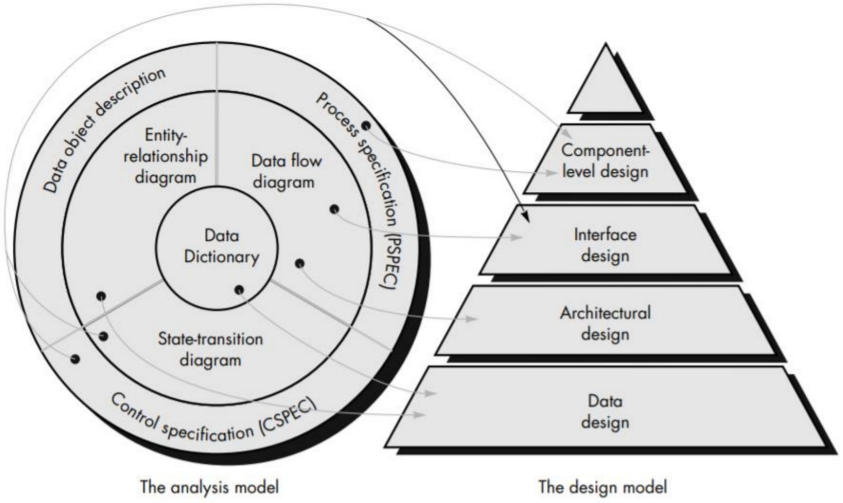


FIGURE 13.1 Translating the analysis model into a software design

Design Model

The data design :
It transforms the information domain model created during analysis into the data structures that will be required to implement the software. The data objects and relationships defined in the entity relationship diagram and the detailed data content depicted in the data dictionary provide the basis for the data design activity. Part of data design may occur in conjunction with the design of software architecture. More detailed data design occurs as each software component is designed

The architectural design :
It defines the relationship between major structural elements of the software, the “design patterns” that can be used to achieve the requirements that have been defined for the system, and the constraints that affect the way in which architectural design patterns can be applied. The architectural design representation—the framework of a computer-based system—can be derived from the system specification, the analysis model, and the interaction of subsystems defined within the analysis model.

Design Model contd…

The interface design :
It describes how the software communicates within itself, with systems that interoperate with it, and with humans who use it. An interface implies a flow of information (e.g., data and/or control) and a specific type of behavior. Therefore, data and control flow diagrams provide much of the information required for interface design.

The component-level design :
It transforms structural elements of the software architecture into a procedural description of software components. Information obtained from the PSPEC, CSPEC, and STD serve as the basis for component design.

During design we make decisions that will ultimately affect the success of software construction and, as important, the ease with which software can be maintained.

Design and Software Quality

Throughout the design process, the quality of the evolving design is assessed with a series of formal technical reviews

Three characteristics that serve as a guide for the evaluation of a good design:

• The design must implement all of the explicit requirements contained in the analysis model, and it must accommodate all of the implicit requirements desired by the customer.

• The design must be a readable, understandable guide for those who generate code and for those who test and subsequently support the software.

• The design should provide a complete picture of the software, addressing the data, functional, and behavioral domains from an implementation perspective

DESIGN PRINCIPLES

Software design is both a process and a model. The *design* process is a sequence of steps that enable the designer to describe all aspects of the software to be built. It is important to note, however, that the design process is not simply a cookbook. Creative skill, past experience, a sense of what makes "good" software, and an overall commitment to quality are critical success factors for a competent design.

Basic design principles enable the software engineer to navigate the design process

Principles for software design

- **The design process should not suffer from "tunnel vision."** A good designer should consider alternative approaches, judging each based on the requirements of the problem, the resources available to do the job, and the design concepts presented
- **The design should be traceable to the analysis model.** Because a single element of the design model often traces to multiple requirements, it is necessary to have a means for tracking how requirements have been satisfied by the design model.
- **The design should not reinvent the wheel.** Systems are constructed using a set of design patterns, many of which have likely been encountered before. These patterns should always be chosen as an alternative to reinvention. Time is short and resources are limited! Design time should be invested in representing truly new ideas and integrating those patterns that already exist.

Principles for Software Design contd..

- **The design should "minimize the intellectual distance" [DAV95] between the software and the problem as it exists in the real world.** That is, the structure of the software design should (whenever possible) mimic the structure of the problem domain.
- **The design should exhibit uniformity and integration.** A design is uniform if it appears that one person developed the entire thing. Rules of style and format should be defined for a design team before design work begins. A design is integrated if care is taken in defining interfaces between design components.
- **The design should be structured to accommodate change.** The design concepts discussed in the next section enable a design to achieve this principle.

- **The design should be structured to degrade gently, even when aberrant data, events, or operating conditions are encountered.** Well-designed software should never "bomb." It should be designed to accommodate unusual circumstances, and if it must terminate processing, do so in a graceful manner.
- **Design is not coding, coding is not design.** Even when detailed procedural designs are created for program components, the level of abstraction of the design model is higher than source code. The only design decisions made at the coding level address the small implementation details that enable the procedural design to be coded.
- **The design should be assessed for quality as it is being created, not after the fact.** A variety of design concepts and design measures are available to assist the designer in assessing quality.
- **The design should be reviewed to minimize conceptual (semantic) errors.** There is sometimes a tendency to focus on minutiae when the design is reviewed, missing the forest for the trees. A design team should ensure that major conceptual elements of the design (omissions, ambiguity, inconsistency) have been addressed before worrying about the syntax of the design model.

Architectural Design

Design is an activity concerned with making major decisions, often of a structural nature. It shares with programming a concern for abstracting information representation and processing sequences, but the level of detail is quite different at the extremes.

Design builds coherent, well planned representations of programs that concentrate on the interrelationships of parts at the higher level and the logical operations involved at the lower levels.

Architectural design represents the structure of data and program components that are required to build a computer-based system.

It considers the architectural style that the system will take, the structure and properties of the components that constitute the system, and the interrelationships that occur among all architectural components of a system

SOFTWARE ARCHITECTURE

Ever since the first program was divided into modules, software systems have had architectures, and programmers have been responsible for the interactions among the modules and the global properties of the assemblage.

Historically, architectures have been implicit— accidents of implementation, or legacy systems of the past. Good software developers have often adopted one or several architectural patterns as strategies for system organization, but they use these patterns informally and have no means to make them explicit in the resulting system.

What Is Architecture?

It is the manner in which the various components of the building(taken as example) are integrated to form a cohesive whole. It is the way in which the building fits into its environment and meshes with other buildings in its vicinity.

It is the degree to which the building meets its stated purpose and satisfies the needs of its owner. It is the aesthetic feel of the structure—the visual impact of the building—and the way textures, colors, and materials are combined to create the external facade and the internal "living environment."

It is small details—the design of lighting fixtures, the type of flooring, the placement of wall hangings, the list is almost endless. And finally, it is art.

What Is Software Architecture?

The software architecture of a program or computing system is the structure or structures of the system, which comprise software components, the externally visible properties of those components, and the relationships among them

What Is Software Architecture? Contd..

The architecture is not the operational software. Rather, it is a representation that enables a software engineer to

(1) analyze the effectiveness of the design in meeting its stated requirements,

(2) consider architectural alternatives at a stage when making design changes is still relatively easy, and

(3) reducing the risks associated with the construction of the software

This definition emphasizes the role of "software components" in any architectural representation. ***In the context of architectural design, a software component can be something as simple as a program module***, but it can also be extended to include databases and "middleware" that enable the configuration of a network of clients and servers. The properties of components are those characteristics that are necessary to an understanding of how the components interact with other components.

Why Is Architecture Important?

There are **three key reasons** that software architecture is important:

- ***Representations of software architecture are an enabler for communication between all parties (stakeholders) interested in the development of a computer-based system.***

- ***The architecture highlights early design decisions that will have a profound impact on all software engineering work that follows and, as important, on the ultimate success of the system as an operational entity.***

- ***Architecture "constitutes a relatively small, intellectually graspable model of how the system is structured and how its components work together".***

ARCHITECTURAL STYLES

The software that is built for computer-based systems also exhibits one of many architectural styles. Each style describes a **system category that encompasses**

(1) a set of components (e.g., a database, computational modules) that perform a function required by a system;

(2) a set of connectors that enable "communication, coordination and cooperation" among components;

(3) constraints that define how components can be integrated to form the system; and

(4) semantic models that enable a designer to understand the overall properties of a system by analyzing the known properties of its constituent parts .

Styles and Patterns of Architectures

1. Data-centered architectures:

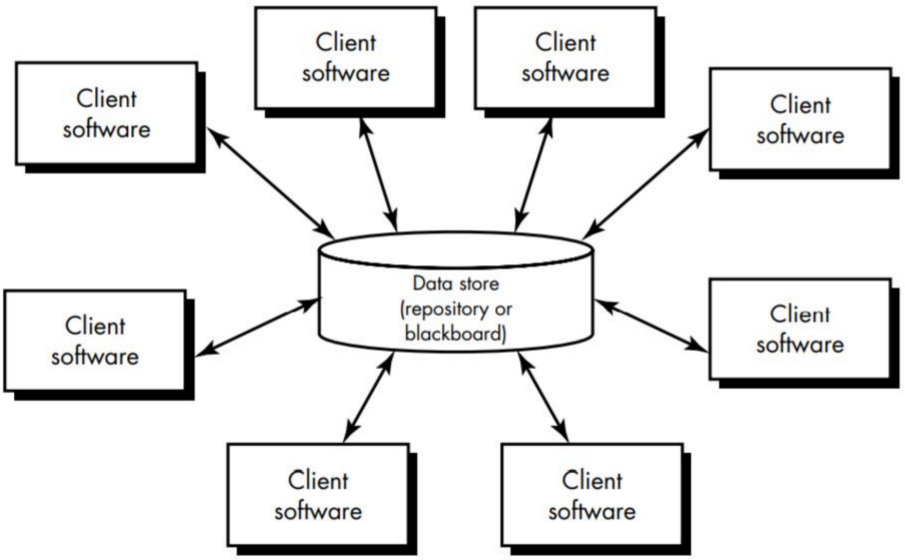


FIGURE 14.1
Data-centered architecture

Data-flow architectures

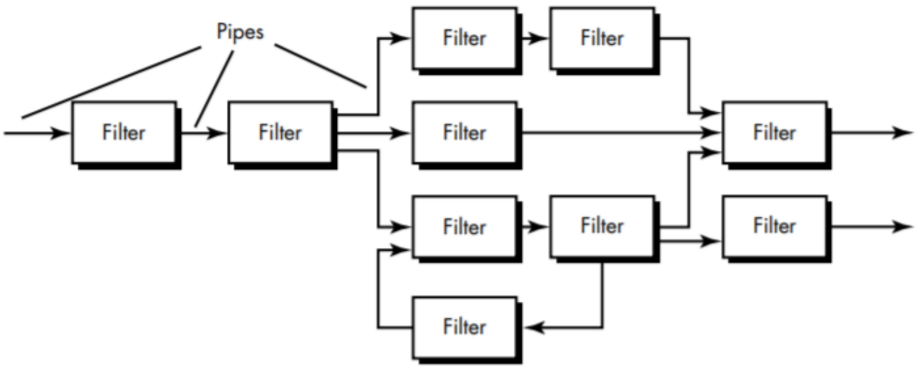
This architecture is applied when input data are to be transformed through a series of computational or manipulative components into output data. A pipe and filter pattern has a set of components, called filters, connected by pipes that transmit data from one component to the next.

Each filter works independently of those components upstream and downstream, is designed to expect data input of a certain form, and produces data output (to the next filter) of a specified form. However, the filter does not require knowledge of the working of its neighboring filters.

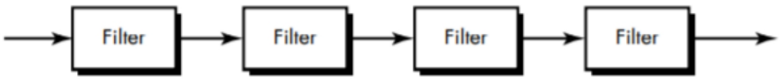
Styles and Patterns of Architectures contd..

2. Data-flow architectures:

FIGURE 14.2
Data flow architectures



(a) Pipes and filters



(b) Batch sequential

Layered architectures

The basic structure of a layered architecture is illustrated in Figure 14.3. A number of different layers are defined, each accomplishing operations that progressively become closer to the machine instruction set. At the outer layer, components service user interface operations. At the inner layer, components perform operating system interfacing. Intermediate layers provide utility services and application software functions.

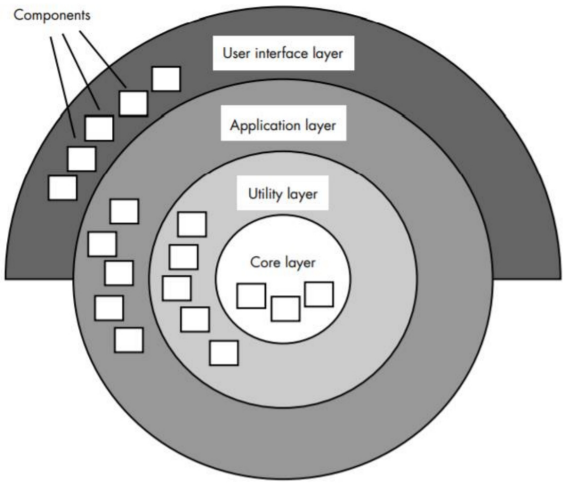


FIGURE 14.3
Layered architecture

Modules

Properties :

i. Well defined subsystem

ii. Well defined purpose

iii. Can be separately compiled and stored in a library.

iv. Module can use other modules

v. Module should be easier to use than to build

vi. Simpler from outside than from the inside.

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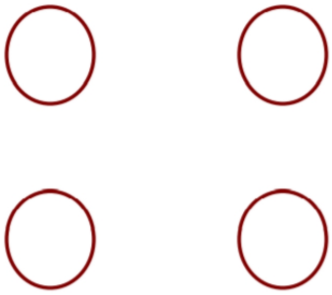
Software Design

Modularity is the single attribute of software that allows a program to be intellectually manageable.

It enhances design clarity, which in turn eases implementation, debugging, testing, documenting, and maintenance of software product.

Module Coupling

Coupling is the measure of the degree of interdependence between modules.



(Uncoupled : no dependencies)

(a)

13.5.3 Coupling

Coupling is a measure of interconnection among modules in a software structure. Coupling depends on the interface complexity between modules, the point at which entry or reference is made to a module, and what data pass across the interface.

In software design, we strive for lowest possible coupling. Simple connectivity among modules results in software that is easier to understand and less prone to a "ripple effect" [STE74], caused when errors occur at one location and propagate through a system.

Figure 13.6 provides examples of different types of module coupling. Modules *a* and *d* are subordinate to different modules. Each is unrelated and therefore no direct coupling occurs. Module *c* is subordinate to module *a* and is accessed via a conventional argument list, through which data are passed. As long as a simple argument list is present (i.e., simple data are passed; a one-to-one correspondence of items exists), low coupling (called *data coupling*) is exhibited in this portion of structure. A variation of data coupling, called *stamp coupling*, is found when a portion of a data structure (rather than simple arguments) is passed via a module interface. This occurs between modules *b* and *a*.

At moderate levels, coupling is characterized by passage of control between modules. *Control coupling* is very common in most software designs and is shown in Figure 13.6 where a "control flag" (a variable that controls decisions in a subordinate or superordinate module) is passed between modules *d* and *e*.

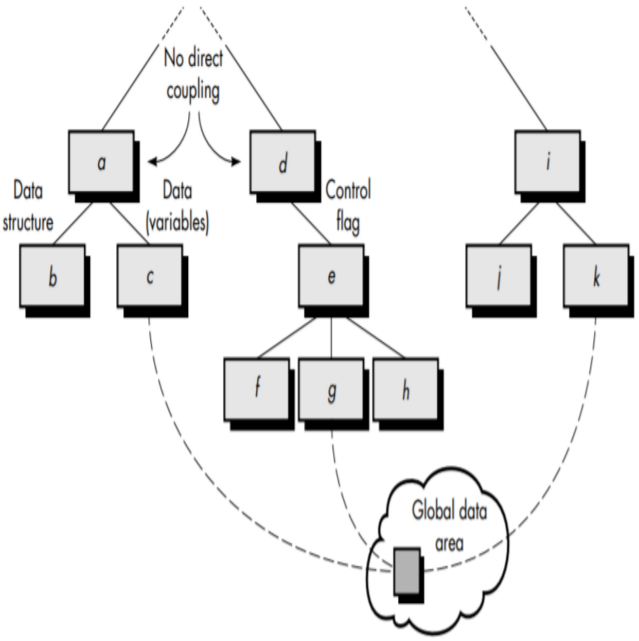


Fig: 13.6- Types of Coupling

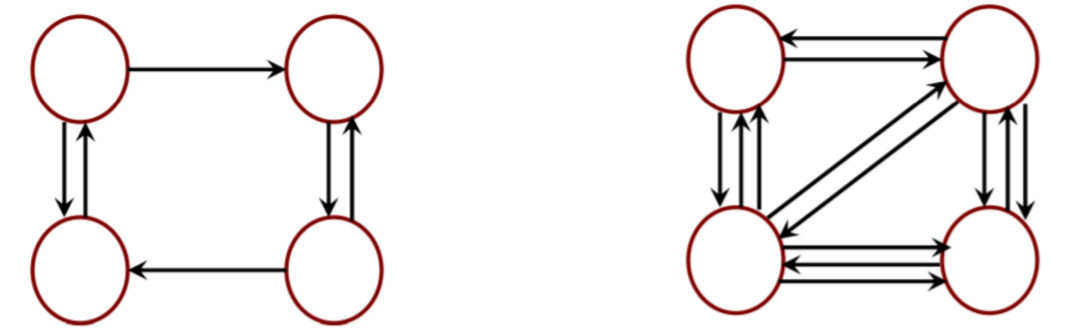


Fig. 5 : Module coupling

This can be achieved as:

- Controlling the number of parameters passed amongst modules.
- Avoid passing undesired data to calling module.
- Maintain parent / child relationship between calling & called modules.
- Pass data, not the control information.

Module Cohesion

Cohesion is a measure of the degree to which the elements of a module are functionally related.

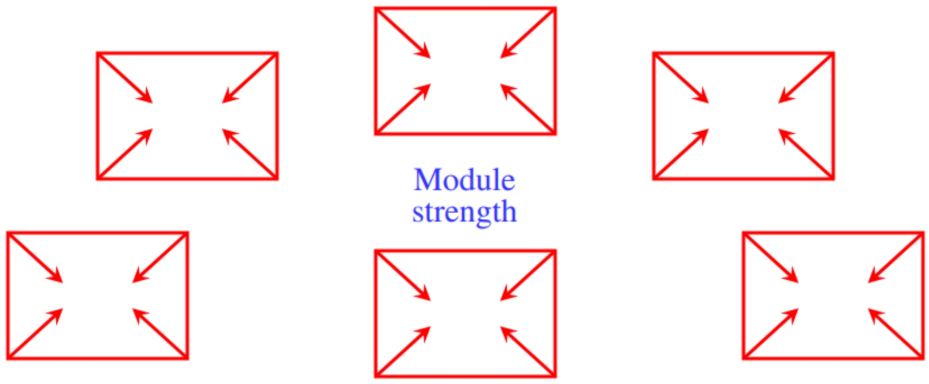


Fig. 10 : Cohesion=Strength of relations within modules

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Types of cohesion

- Functional cohesion
- Sequential cohesion
- Procedural cohesion
- Temporal cohesion
- Logical cohesion
- Coincident cohesion

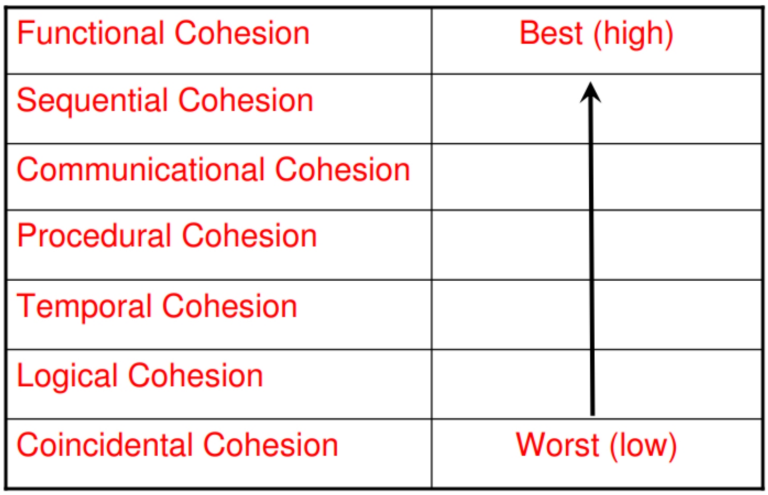

Fig. 11 : Types of module cohesion

Functional Cohesion

- A and B are part of a single functional task. This is very good reason for them to be contained in the same procedure.

Sequential Cohesion

- Module A outputs some data which forms the input to B. This is the reason for them to be contained in the same procedure.

Procedural Cohesion

- Procedural Cohesion occurs in modules whose instructions although accomplish different tasks yet have been combined because there is a specific order in which the tasks are to be completed.

Temporal Cohesion

- Module exhibits temporal cohesion when it contains tasks that are related by the fact that all tasks must be executed in the same time-span.

Logical Cohesion

- Logical cohesion occurs in modules that contain instructions that appear to be related because they fall into the same logical class of functions.

Coincidental Cohesion

- Coincidental cohesion exists in modules that contain instructions that have little or no relationship to one another.

Relationship between Cohesion & Coupling

If the software is not properly modularized, a host of seemingly trivial enhancement or changes will result into death of the project. Therefore, a software engineer must design the modules with goal of high cohesion and low coupling.

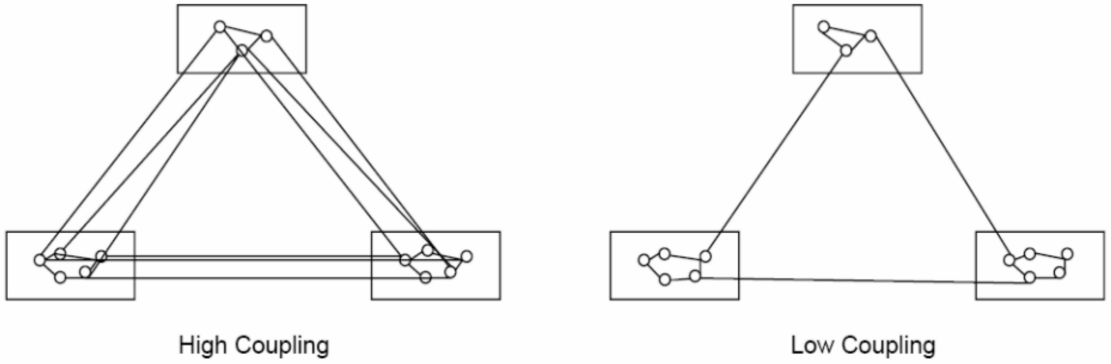

Fig. 12 : View of cohesion and coupling

STRATEGY OF DESIGN

A good system design strategy is to organize the program modules in such a way that are easy to develop and latter to, change. Structured design techniques help developers to deal with the size and complexity of programs. Analysts create instructions for the developers about how code should be written and how pieces of code should fit together to form a program. It is important for two reasons:

- First, even pre-existing code, if any, needs to be understood, organized and pieced together.
- Second, it is still common for the project team to have to write some code and produce original programs that support the application logic of the system.

Bottom-Up Design

These modules are collected together in the form of a "library".

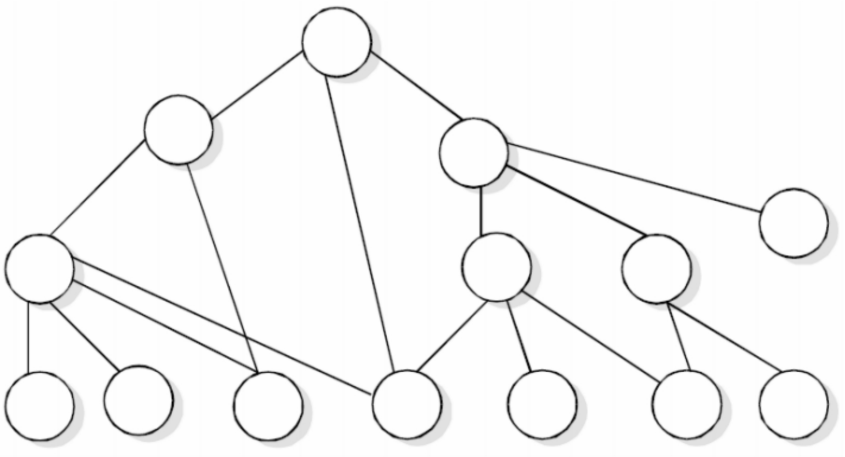

Fig. 13 : Bottom-up tree structure

Top-Down Design

A top down design approach starts by identifying the major modules of the system, decomposing them into their lower level modules and iterating until the desired level of detail is achieved. This is stepwise refinement; starting from an abstract design, in each step the design is refined to a more concrete level, until we reach a level where no more refinement is needed and the design can be implemented directly.

Hybrid Design

For top-down approach to be effective, some bottom-up approach is essential for the following reasons:

- To permit common sub modules.
- Near the bottom of the hierarchy, where the intuition is simpler, and the need for bottom-up testing is greater, because there are more number of modules at low levels than high levels.
- In the use of pre-written library modules, in particular, reuse of modules.

FUNCTION ORIENTED DESIGN

Function Oriented design is an approach to software design where the design is decomposed into a set of interacting units where each unit has a clearly defined function. Thus, system is designed from a functional viewpoint.

If a program is created top-down, the modules become very specialized. As one can easily see in top down design structure, each module is used by at most one other module, its parent. For a module, however, we must require that several other modules as in design reusable structure as shown in fig. 15.

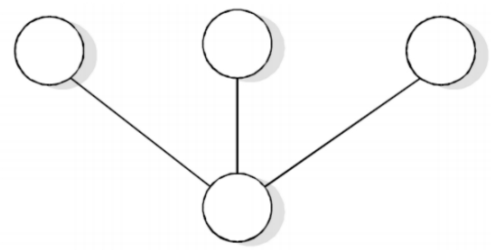

Fig. 15 : Design reusable structure

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Design Notations

Design notations are largely meant to be used during the process of design and are used to represent design or design decisions. For a function oriented design, the design can be represented graphically or mathematically by the following:

- Data flow diagrams
- Data Dictionaries
- Structure Charts
- Pseudocode

Software Metrics

Software Metrics: What and Why ?

1. How to measure the size of a software?
2. How much will it cost to develop a software?
3. How many bugs can we expect?
4. When can we stop testing?
5. When can we release the software?

Software Metrics

6. What is the complexity of a module?
7. What is the module strength and coupling?
8. What is the reliability at the time of release?
9. Which test technique is more effective?
10. Are we testing hard or are we testing smart?
11. Do we have a strong program or a week test suite?

Software Metrics

- Pressman explained as “A measure provides a quantitative indication of the extent, amount, dimension, capacity, or size of some attribute of the product or process”.
- Measurement is the act of determine a measure
- The metric is a quantitative measure of the degree to which a system, component, or process possesses a given attribute.
- Fenton defined measurement as “ it is the process by which numbers or symbols are assigned to attributes of entities in the real world in such a way as to describe them according to clearly defined rules”.

Software Metrics

- Definition

Software metrics can be defined as "*The continuous application of measurement based techniques to the software development process and its products to supply meaningful and timely management information, together with the use of those techniques to improve that process and its products*".

Structure Chart

It partition a system into block boxes. A black box means that functionality is known to the user without the knowledge of internal design.

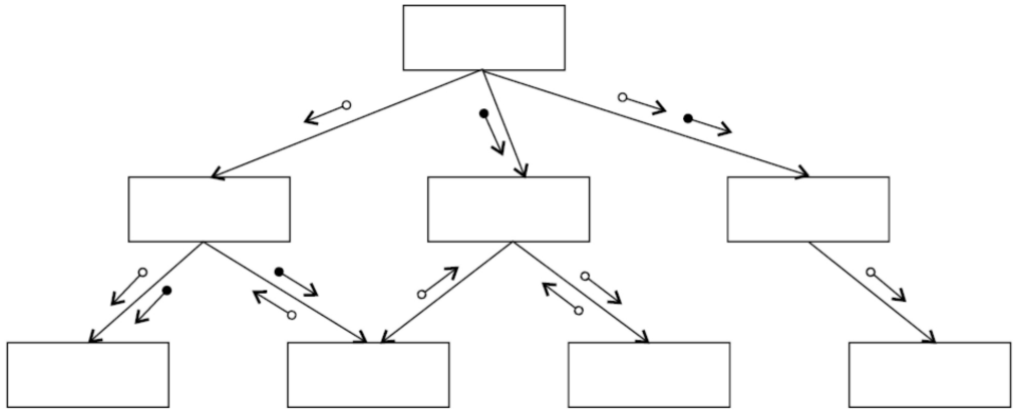

Fig. 16 : Hierarchical format of a structure chart

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A transaction centered structure describes a system that processes a number of different types of transactions. It is illustrated in Fig.19.

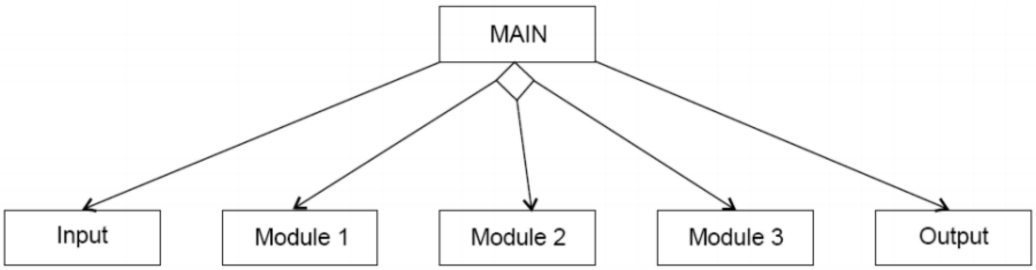

Fig. 19 : Transaction-centered structure

In the above figure the MAIN module controls the system operation its functions is to:

- invoke the INPUT module to read a transaction;
- determine the kind of transaction and select one of a number of transaction modules to process that transaction, and
- output the results of the processing by calling OUTPUT module.

Pseudocode

Pseudocode notation can be used in both the preliminary and detailed design phases.

Using pseudocode, the designer describes system characteristics using short, concise, English language phrases that are structured by key words such as It-Then-Else, While-Do, and End.

Software Design

- Definitions

i. **Design entity.** An element (Component) of a design that is structurally and functionally distinct from other elements and that is separately named and referenced.

ii. **Design View.** A subset of design entity attribute information that is specifically suited to the needs of a software project activity.

iii. **Entity attributes.** A named property or characteristics of a design entity. It provides a statement of fact about the entity.

iv. **Software design description (SDD).** A representation of a software system created to facilitate analysis, planning, implementation and decision making.

Software Design

- Purpose of an SDD

The SDD shows how the software system will be structured to satisfy the requirements identified in the SRS. It is basically the translation of requirements into a description of the software structure, software components, interfaces, and data necessary for the implementation phase. Hence, SDD becomes the blue print for the implementation activity.

- Design Description Information Content
  - Introduction
  - Design entities
  - Design entity attributes

Software Design

Object Oriented Design

Object oriented design is the result of focusing attention not on the function performed by the program, but instead on the data that are to do manipulated by the program. Thus, it is orthogonal to function oriented design.

Object Oriented Design begins with an examination of the real world “things” that are part of the problem to be solved. These things (which we will call objects) are characterized individually in terms of their attributes and behavior.

Software Design

➢ Basic Concepts

Object Oriented Design is not dependent on any specific implementation language. Problems are modeled using objects. Objects have:

- Behavior (they do things)
- State (which changes when they do things)

The various terms related to object design are:

i. Objects

The word “Object” is used very frequently and conveys different meaning in different circumstances. Here, meaning is an entity able to save a state (information) and which offers a number of operations (behavior) to either examine or affect this state. An object is characterized by number of operations and a state which remembers the effect of these operations.

ii. Messages

Objects communicate by message passing. Messages consist of the identity of the target object, the name of the requested operation and any other operation needed to perform the function. Message are often implemented as procedure or function calls.

iii. Abstraction

In object oriented design, complexity is managed using abstraction. Abstraction is the elimination of the irrelevant and the amplification of the essentials.

iv. Class

In any system, there shall be number of objects. Some of the objects may have common characteristics and we can group the objects according to these characteristics. This type of grouping is known as a class. Hence, a class is a set of objects that share a common structure and a common behavior.

We may define a class “car” and each object that represent a car becomes an instance of this class. In this class “car”, Indica, Santro, Maruti, Indigo are instances of this class as shown in fig. 20.

Classes are useful because they act as a blueprint for objects. If we want a new square we may use the square class and simply fill in the particular details (i.e. colour and position) fig. 21 shows how can we represent the square class.

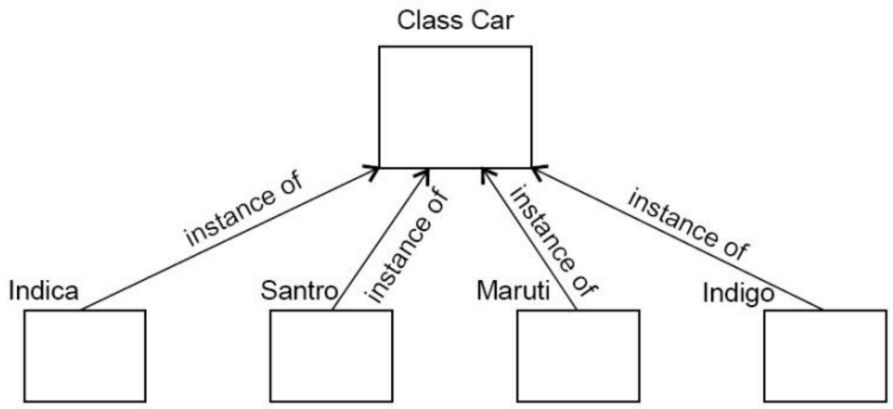

Fig.20: Indica, Santro, Maruti, Indigo are all instances of the class "car"

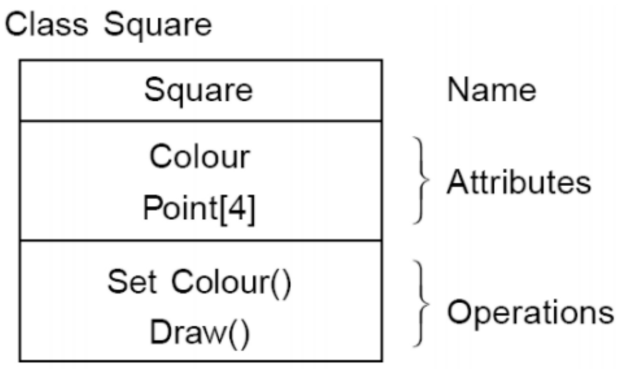

Fig. 21: The square class

v. Attributes

An attributes is a data value held by the objects in a class. The square class has two attributes: a colour and array of points. Each attributes has a value for each object instance. The attributes are shown as second part of the class as shown in fig. 21.

vi. Operations

An operation is a function or transformation that may be applied to or by objects in a class. In the square class, we have two operations: set colour() and draw(). All objects in a class share the same operations. An object “knows” its class, and hence the right implementation of the operation. Operation are shown in the third part of the class as indicated in fig. 21.

vii. Inheritance

Imagine that, as well as squares, we have triangle class. Fig. 22 shows the class for a triangle.

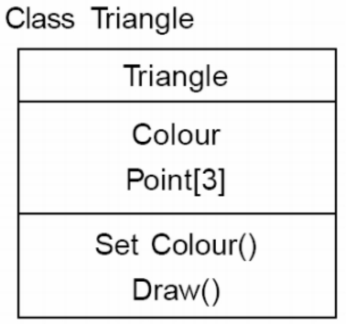

Fig. 22: The triangle class

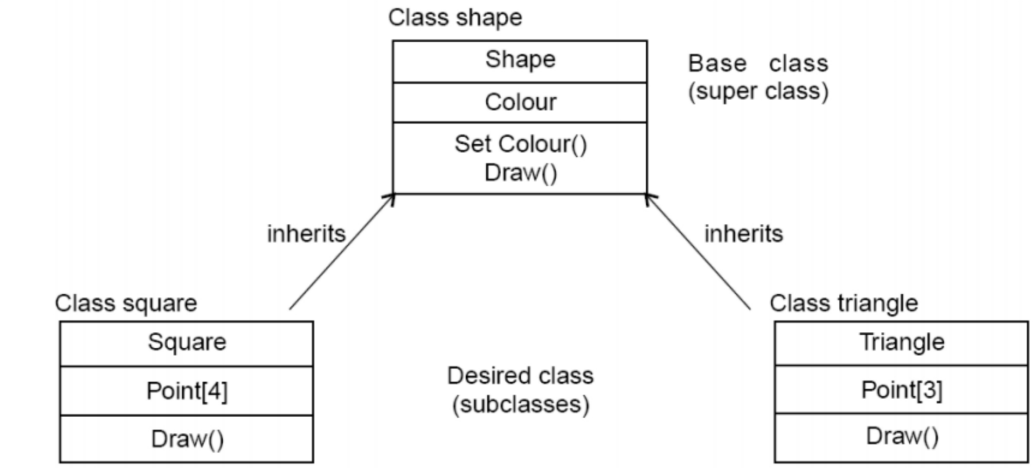

Fig. 23: Abstracting common features in a new class

This sort of abstraction is called inheritance. The low level classes (known as subclasses or derived classes) inherit state and behavior from this high level class (known as a super class or base class).

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viii. Polymorphism

When we abstract just the interface of an operation and leave the implementation to subclasses it is called a polymorphic operation and process is called polymorphism.

ix. Encapsulation (Information Hiding)

Encapsulation is also commonly referred to as “Information Hiding”. It consists of the separation of the external aspects of an object from the internal implementation details of the object.

x. Hierarchy

Hierarchy involves organizing something according to some particular order or rank. It is another mechanism for reducing the complexity of software by being able to treat and express sub-types in a generic way.

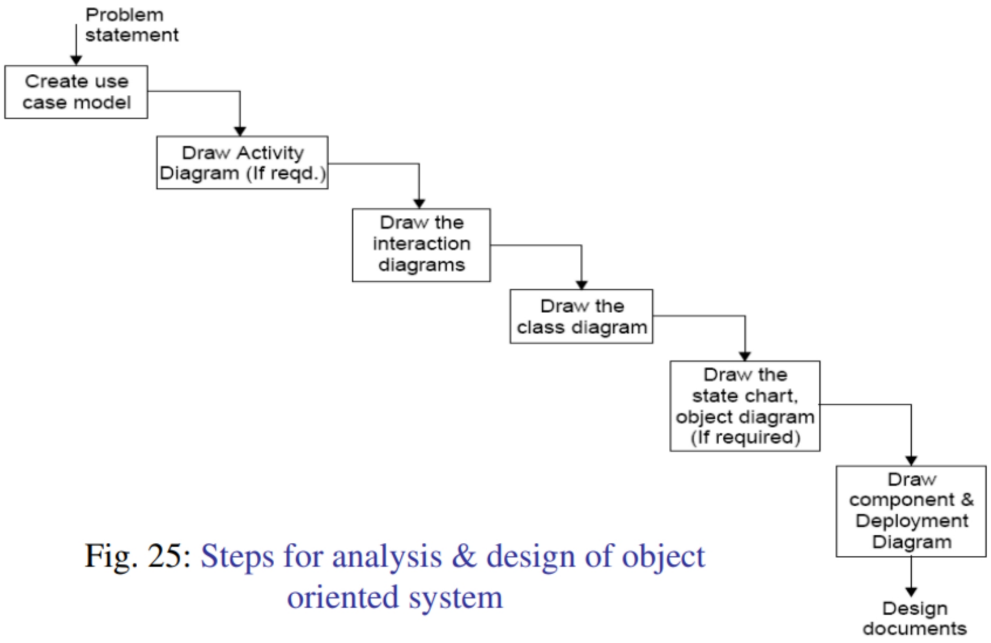
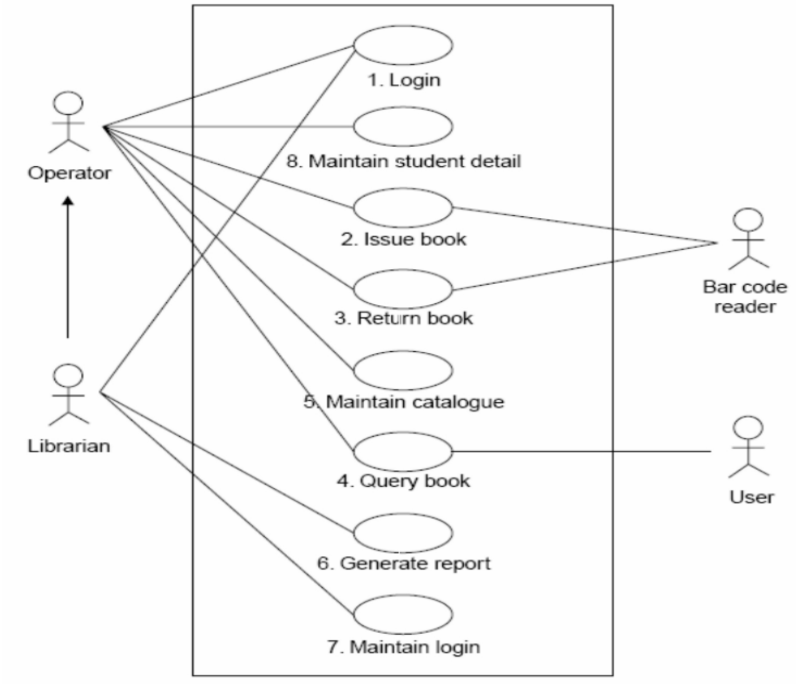


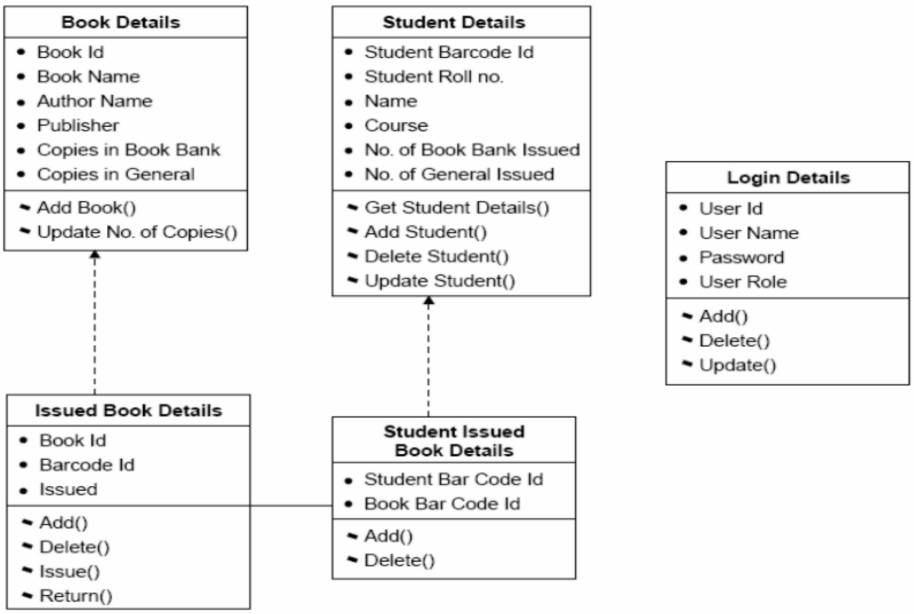
Fig. 25: Steps for analysis & design of object oriented system



Use case diagram for library management system

Software Design

Class diagram of entity classes



Class diagram of entity classes

3

Software Metrics

Software Metrics: What and Why ?

1. How to measure the size of a software?
2. How much will it cost to develop a software?
3. How many bugs can we expect?
4. When can we stop testing?
5. When can we release the software?

19.1.1 McCall's Quality Factors

The factors that affect software quality can be categorized in two broad groups: (1) factors that can be directly measured (e.g., defects per function-point) and (2) factors that can be measured only indirectly (e.g., usability or maintainability). In each case measurement must occur. We must compare the software (documents, programs, data) to some datum and arrive at an indication of quality.

McCall, Richards, and Walters [MCC77] propose a useful categorization of factors that affect software quality. These software quality factors, shown in Figure 19.1, focus on three important aspects of a software product: its operational characteristics, its ability to undergo change, and its adaptability to new environments.

Referring to the factors noted in Figure 19.1, McCall and his colleagues provide the following descriptions:

Correctness. The extent to which a program satisfies its specification and fulfills the customer's mission objectives.

Reliability. The extent to which a program can be expected to perform its intended function with required precision.

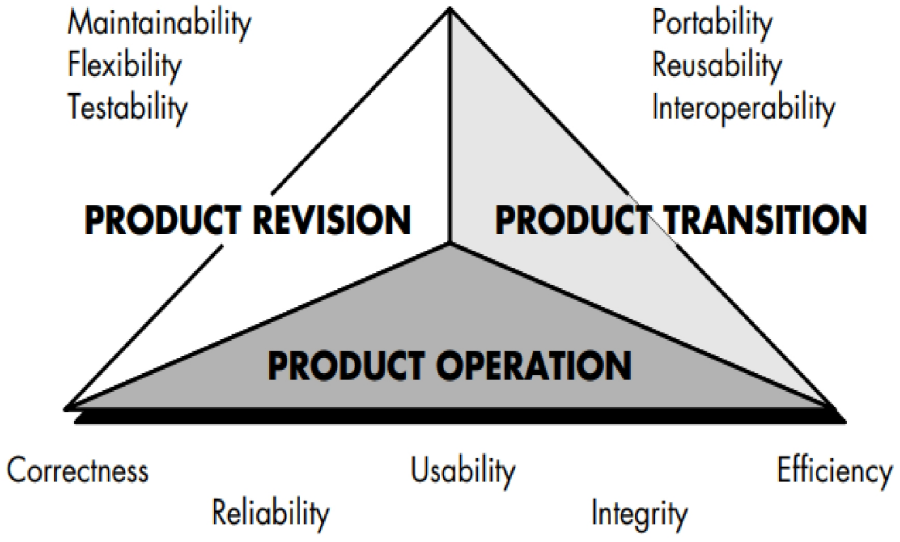


FIGURE 19.1
McCall's software quality factors

Efficiency. The amount of computing resources and code required by a program to perform its function.

Integrity. Extent to which access to software or data by unauthorized persons can be controlled.

Usability. Effort required to learn, operate, prepare input, and interpret output of a program.

Maintainability. Effort required to locate and fix an error in a program. [This is a very limited definition.]

Flexibility. Effort required to modify an operational program.

Testability. Effort required to test a program to ensure that it performs its intended function.

Portability. Effort required to transfer the program from one hardware and/or software system environment to another.

Reusability. Extent to which a program [or parts of a program] can be reused in other applications—related to the packaging and scope of the functions that the program performs.

Interoperability. Effort required to couple one system to another.

Auditability. The ease with which conformance to standards can be checked.

Accuracy. The precision of computations and control.

Communication commonality. The degree to which standard interfaces, protocols, and bandwidth are used.

Completeness. The degree to which full implementation of required function has been achieved.

Conciseness. The compactness of the program in terms of lines of code.

Consistency. The use of uniform design and documentation techniques throughout the software development project.

Data commonality. The use of standard data structures and types throughout the program.

Error tolerance. The damage that occurs when the program encounters an error.

Execution efficiency. The run-time performance of a program.

Expandability. The degree to which architectural, data, or procedural design can be extended.

Generality. The breadth of potential application of program components.

Hardware independence. The degree to which the software is decoupled from the hardware on which it operates.

Instrumentation. The degree to which the program monitors its own operation and identifies errors that do occur.

Modularity. The functional independence of program components.

Operability. The ease of operation of a program.

Security. The availability of mechanisms that control or protect programs and data.

Self-documentation. The degree to which the source code provides meaningful documentation.

Simplicity. The degree to which a program can be understood without difficulty.

Software system independence. The degree to which the program is independent of nonstandard programming language features, operating system characteristics, and other environmental constraints.

Traceability. The ability to trace a design representation or actual program component back to requirements.

Training. The degree to which the software assists in enabling new users to apply the system.

FIGURE 19.2
Quality factors and metrics

Quality factor \ Software quality metric	Correctness	Reliability	Efficiency	Integrity	Maintainability	Flexibility	Testability	Portability	Reusability	Interoperability	Usability
Auditability				x			x				
Accuracy		x									
Communication commonality										x	
Completeness	x										
Complexity		x				x	x				
Concision			x		x	x					
Consistency	x	x			x	x					
Data commonality										x	
Error tolerance		x									
Execution efficiency			x								
Expandability						x					
Generality						x		x	x	x	
Hardware Indep.								x	x		
Instrumentation				x	x		x				
Modularity		x			x	x	x	x	x	x	
Operability			x								x
Security				x							
Self-documentation					x	x	x	x	x		
Simplicity		x			x	x	x				
System Indep.								x	x		
Traceability	x										
Training											x

(Adapted from Arthur, L. A., *Measuring Programmer Productivity and Software Quality*, Wiley-Interscience, 1985.)

functionality, usability, reliability, performance, and supportability. The FURPS quality factors draw liberally from earlier work, defining the following attributes for each of the five major factors:

- *Functionality* is assessed by evaluating the feature set and capabilities of the program, the generality of the functions that are delivered, and the security of the overall system.
- *Usability* is assessed by considering human factors , overall aesthetics, consistency, and documentation.
- *Reliability* is evaluated by measuring the frequency and severity of failure, the accuracy of output results, the mean-time-to-failure (MTTF), the ability to recover from failure, and the predictability of the program.
- *Performance* is measured by processing speed, response time, resource consumption, throughput, and efficiency.
- *Supportability* combines the ability to extend the program (extensibility), adaptability, serviceability—these three attributes represent a more common term, maintainability—in addition, testability, compatibility, configurability (the ability to organize and control elements of the software configuration, , the ease with which a system can be installed, and the ease with which problems can be localized.

METRICS FOR SOURCE CODE

Halstead's theory of software science [HAL77] is one of "the best known and most thoroughly studied . . . composite measures of (software) complexity" [CUR80]. Software science proposed the first analytical "laws" for computer software.[9]

Software science assigns quantitative laws to the development of computer software, using a set of primitive measures that may be derived after code is generated or estimated once design is complete. These follow:

n_1 = the number of distinct operators that appear in a program.
n_2 = the number of distinct operands that appear in a program.
N_1 = the total number of operator occurrences.
N_2 = the total number of operand occurrences.

Halstead uses these primitive measures to develop expressions for the overall *program length, potential minimum volume* for an algorithm, the *actual volume* (number of bits required to specify a program), the *program level* (a measure of software complexity), the *language level* (a constant for a given language), and other features such as development effort, development time, and even the projected number of faults in the software.

Halstead shows that length N can be estimated

$$N = n_1 \log_2 n_1 + n_2 \log_2 n_2$$

and program volume may be defined

$$V = N \log_2 (n_1 + n_2)$$

It should be noted that V will vary with programming language and represents the volume of information (in bits) required to specify a program.

Theoretically, a minimum volume must exist for a particular algorithm. Halstead defines a volume ratio L as the ratio of volume of the most compact form of a program to the volume of the actual program. In actuality, L must always be less than 1. In terms of primitive measures, the volume ratio may be expressed as

$$L = 2/n_1 \times n_2/N_2$$

Software Project Planning

Software planning begins before technical work starts, continues as the software evolves from concept to reality, and culminates only when the software is retired.

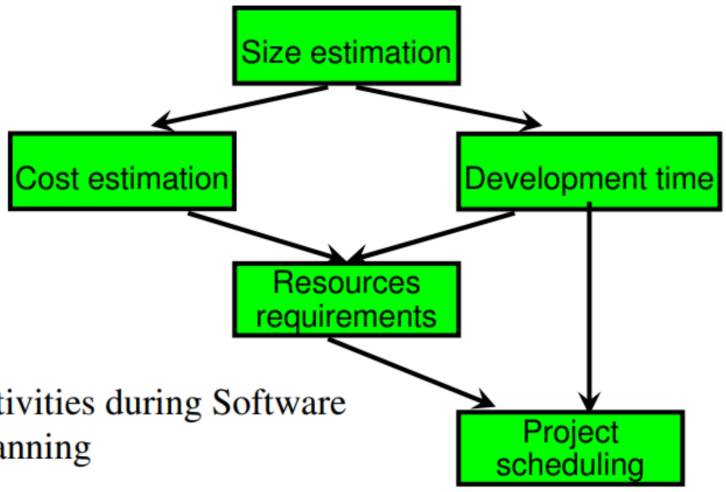


Fig. 1: Activities during Software Project Planning

"A line of code is any line of program text that is not a comment or blank line, regardless of the number of statements or fragments of statements on the line. This specifically includes all lines containing program header, declaration, and executable and non-executable statements".

Function Count

Alan Albrecht while working for IBM, recognized the problem in size measurement in the 1970s, and developed a technique (which he called Function Point Analysis), which appeared to be a solution to the size measurement problem.

Software Project Planning

The principle of Albrecht's function point analysis (FPA) is that a system is decomposed into functional units.

- Inputs : information entering the system
- Outputs : information leaving the system
- Enquiries : requests for instant access to information
- Internal logical files : information held within the system
- External interface files : information held by other system that is used by the system being analyzed.

The FPA functional units are shown in figure given below:

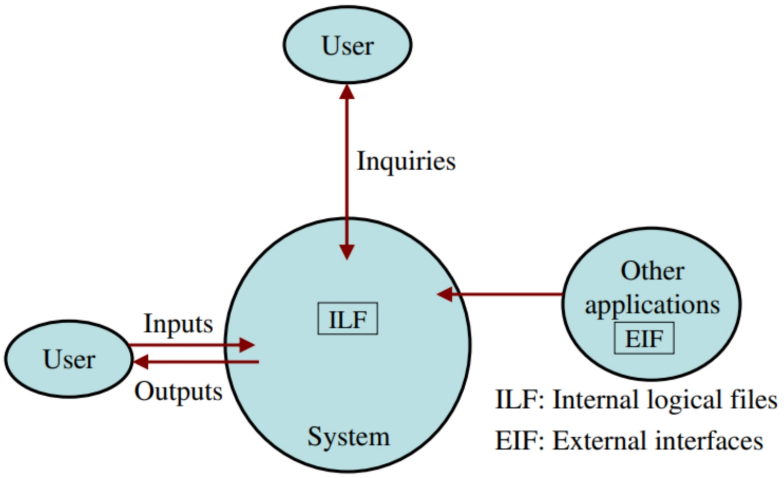


Fig. 3: FPAs functional units System

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The five functional units are divided in two categories:

(i) Data function types

- Internal Logical Files (ILF): A user identifiable group of logical related data or control information maintained within the system.
- External Interface files (EIF): A user identifiable group of logically related data or control information referenced by the system, but maintained within another system. This means that EIF counted for one system, may be an ILF in another system.

(ii) Transactional function types

- External Input (EI): An EI processes data or control information that comes from outside the system. The EI is an elementary process, which is the smallest unit of activity that is meaningful to the end user in the business.
- External Output (EO): An EO is an elementary process that generate data or control information to be sent outside the system.
- External Inquiry (EQ): An EQ is an elementary process that is made up to an input-output combination that results in data retrieval.

Special features

- Function point approach is independent of the language, tools, or methodologies used for implementation; i.e. they do not take into consideration programming languages, data base management systems, processing hardware or any other data base technology.
- Function points can be estimated from requirement specification or design specification, thus making it possible to estimate development efforts in early phases of development.

- Function points are directly linked to the statement of requirements; any change of requirements can easily be followed by a re-estimate.
- Function points are based on the system user's external view of the system, non-technical users of the software system have a better understanding of what function points are measuring.

Functions points may compute the following important metrics:

Productivity	=	FP / persons-months
Quality	=	Defects / FP
Cost	=	Rupees / FP
Documentation	=	Pages of documentation per FP

These metrics are controversial and are not universally acceptable. There are standards issued by the International Functions Point User Group (IFPUG, covering the Albrecht method) and the United Kingdom Function Point User Group (UFPGU, covering the MK11 method). An ISO standard for function point method is also being developed.

BASIS PATH TESTING

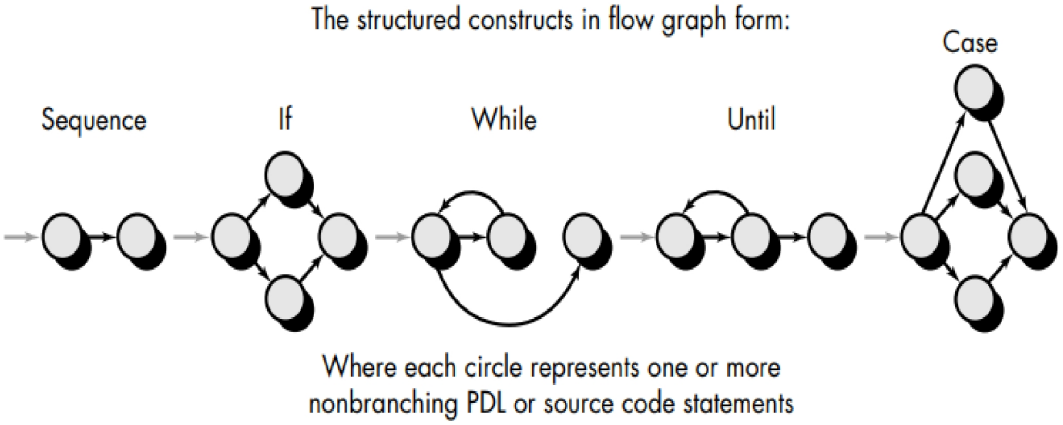
Basis path testing is a white-box testing technique first proposed by Tom McCabe [MCC76]. The basis path method enables the test case designer to derive a logical complexity measure of a procedural design and use this measure as a guide for defining a basis set of execution paths. Test cases derived to exercise the basis set are guaranteed to execute every statement in the program at least one time during testing.

17.4.1 Flow Graph Notation

Before the basis path method can be introduced, a simple notation for the representation of control flow, called a *flow graph* (or *program graph*) must be introduced.[3] The flow graph depicts logical control flow using the notation illustrated in Figure 17.1. Each structured construct has a corresponding flow graph symbol.

To illustrate the use of a flow graph, we consider the procedural design representation in Figure 17.2A. Here, a flowchart is used to depict program control structure.

FIGURE 17.1
Flow graph notation



17.4.2 Cyclomatic Complexity

Cyclomatic complexity is a software metric that provides a quantitative measure of the logical complexity of a program. When used in the context of the basis path testing method, the value computed for cyclomatic complexity defines the number of independent paths in the basis set of a program and provides us with an upper bound for the number of tests that must be conducted to ensure that all statements have been executed at least once.

An *independent path* is any path through the program that introduces at least one new set of processing statements or a new condition. When stated in terms of a flow

FIGURE 17.2
Flowchart, (A) and flow graph (B)

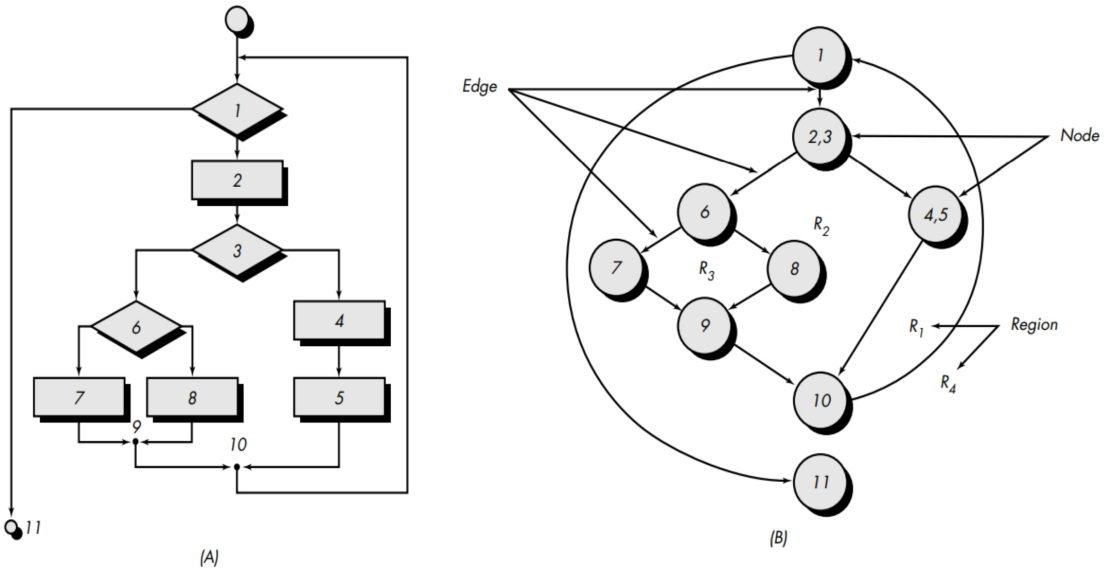
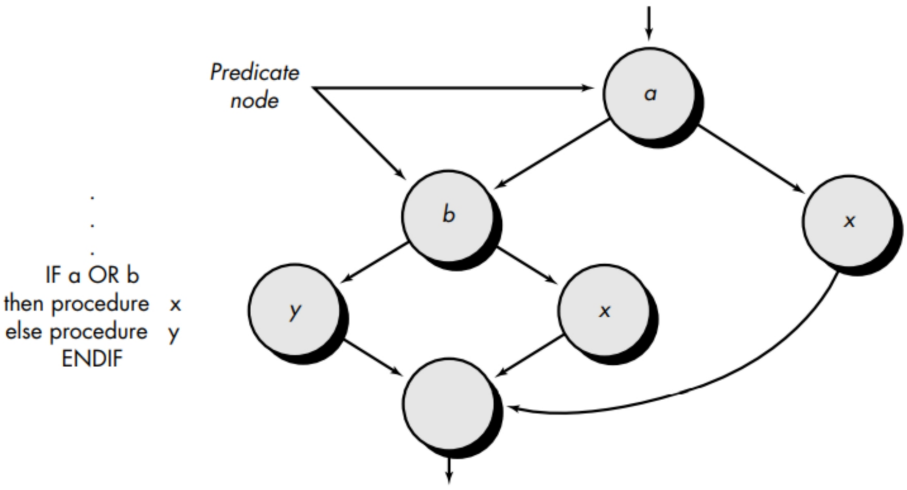


FIGURE 17.3
Compound logic



graph, an independent path must move along at least one edge that has not been traversed before the path is defined. For example, a set of independent paths for the flow graph illustrated in Figure 17.2B is

Cyclomatic complexity has a foundation in graph theory and provides us with an extremely useful software metric. Complexity is computed in one of three ways:

1. The number of regions of the flow graph correspond to the cyclomatic complexity.
2. Cyclomatic complexity, $V(G)$, for a flow graph, G, is defined as

   $$V(G) = E - N + 2$$

   where E is the number of flow graph edges, N is the number of flow graph nodes.
3. Cyclomatic complexity, $V(G)$, for a flow graph, G, is also defined as

   $$V(G) = P + 1$$

   where P is the number of predicate nodes contained in the flow graph G.

where E is the number of flow graph edges, N is the number of flow graph nodes.

3. Cyclomatic complexity, $V(G)$, for a flow graph, G, is also defined as

$$V(G) = P + 1$$

where P is the number of predicate nodes contained in the flow graph G.

Referring once more to the flow graph in Figure 17.2B, the cyclomatic complexity can be computed using each of the algorithms just noted:

1. The flow graph has four regions.
2. $V(G) = 11$ edges $- 9$ nodes $+ 2 = 4$.
3. $V(G) = 3$ predicate nodes $+ 1 = 4$.

Therefore, the cyclomatic complexity of the flow graph in Figure 17.2B is 4.

More important, the value for $V(G)$ provides us with an upper bound for the number of independent paths that form the basis set and, by implication, an upper bound on the number of tests that must be designed and executed to guarantee coverage of all program statements.

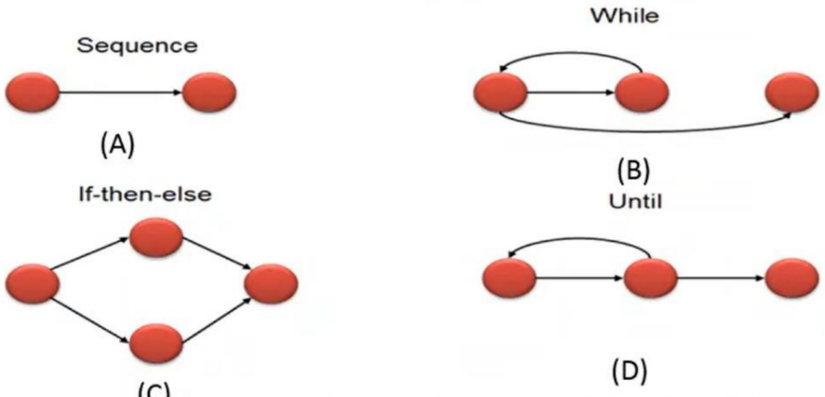


Figure: CFG for normal sequence of flow, While, if-else and until.

Method 1

V(G)= E-N+2

(A) V(G)= 1-2+2=1
(B) V(G)= 3-3+2=2
(C) V(G)= 4-4+2=2
(D) V(G)= 3-3+2=2

Method 2

V(G)=Total no. of bounded areas+1

(A) V(G)= 0+1=1
(B) V(G)= 1+1=2
(C) V(G)= 1+1=2
(D) V(G)= 1+1=2

Method 3

V(G)=P+1

(A) V(G)= 0+1=1
(B) V(G)= 1+1=2
(C) V(G)= 1+1=2
(D) V(G)= 1+1=2

Where P = No. of predicate nodes (node that contains condition)